2.3 Friction, Angled Forces and Tension

Learning Goals (Students will be able too…)

- Solve problems with objects on inclined planes, with friction involved.
- Draw FBD's and break up forces into their components.
- Compare static and kinetic friction.
- Make the connection between the steepness of an incline and the force of friction.

Static vs. Kinetic Friction

- **Static Friction:** friction that occurs between two solid objects that are <u>not moving</u> relative to each other.
  - Acts to prevent motion completely.
- **Coefficient of Static Friction:** $F_S = \mu_S F_N$
- **Kinetic Friction (also known as sliding friction):** friction that occurs when two solid objects are <u>moving</u> relative to each other.
  - Acts to slow down motion.
- **Coefficient of Kinetic Friction:** $F_K = \mu_K F_N$

Static and Kinetic Friction on an Inclined Plane -Simulation

The following simulation is located at: <u>https://ophysics.com/f2.html</u>

1) What happens to the kinetic friction when you set the static friction to 0? Why might this be?

The kinetic friction drops to 0 because this simulation constrains $\mu_k \leq \mu_s$.

2) If we set kinetic friction to 0 but we have some static friction of some sort, what happens?

The static friction holds the box in place while the ramp is not too steep, but once the ramp is steep enough the box starts accelerating as if the surfnce is frictionless.

3) Notice that the static friction in this simulation will not exceed 1. There are cases though in real life where friction exceeds the coefficient of 1 though? Why do you think they restricted it?

Most everyday surfaces have coefficients less than 1. Values above 1 are special cases, like rubber on dry pavement.

4) Kinetic Friction is based on static friction. Notice how kinetic can never exceed static, why might this be?

At the moment it starts to slide, the push equals $\mu_s F_N$. If kinetic friction were bigger, $F_{net} = (\mu_s - \mu_k) F_N$ would be negative, so the object would slow down and stop as soon as it started. Objects keep sliding, so $\mu_k \leq \mu_s$.

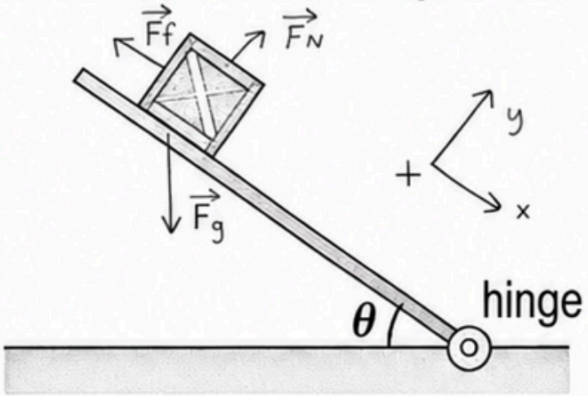
Static or Kinetic?

Example 1) Identify the following as an example of static or kinetic friction (put an S for static a K for kinetic)

a. a sled moving along the snow K
b. a fridge that you push on but it won't move S
c. the friction between a car's tires and the road as it drives along S
d. the friction between a car's tires and the road as skids to a stop K

Friction Problems - Classroom Examples

Example 1) Overcoming Static Friction A crate is placed on an inclined board as shown. One end of the board is hinged so that the angle θ is adjustable. The coefficient of static friction between the crate and the board is **0.30**. Determine the angle at which the crate just begins to slip.



$\mu_s = 0.3 \quad a_x = 0 \quad a_y = 0 \quad A = ?$

$F_{gx} = mg \sin A$

$F_{sx} = -\mu mg \cos A$

$F_{netx} = ma_x$

$F_{netx} = 0$

$0 = F_{gx} + F_{sx}$

$0 = mg \sin A - \mu mg \cos A$

$\frac{\mu mg \cos A = mg \sin A}{mg}$

$\mu \cos A = \sin A$

$\mu = \frac{\sin A}{\cos A}$

$\mu = \tan A$

$\tan^{-1}(\mu) = A$

$\tan^{-1}(0.30) = A$

$16.699^\circ = A$

$A = 17^\circ$

$F_{gy} = -mg \cos A$

$F_{nety} = ma_y$

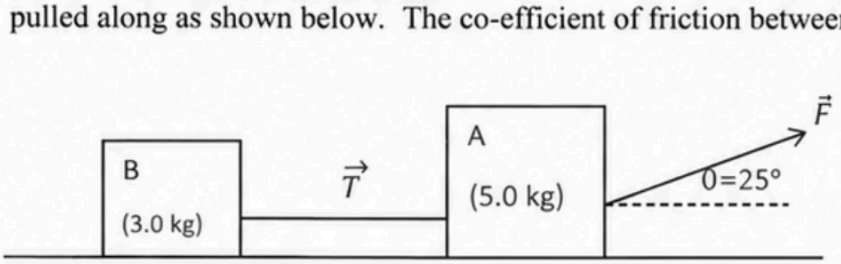
$F_{nety} = 0$

$F_{nety} = F_N + F_{gy}$

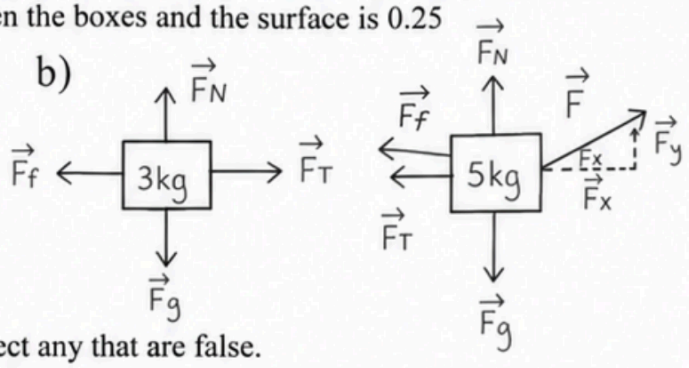
$0 = F_N - mg \cos A$

$F_N = +mg \cos A$

Example 2) Two Body Problems Two boxes are attached by a light, inextensible (not stretchy) string and pulled along as shown below. The co-efficient of friction between the boxes and the surface is 0.25



b)



a) State whether each of the following is true or false. Correct any that are false.

a. The x component of F is equal to the magnitude of T

False: the x component of $\vec{F}$ is greater than the magnitade of $\vec{T}$

b. The acceleration of both boxes are equal

True

c. the net force on both boxes are equal

False: $F_{net} = ma$.

$F_{net,A} > F_{net,B}$ (equal only if $a = 0$)

d. All forces acting on box B are either vertical or horizontal

True

b) Draw a free body diagram of each box (draw in the space beside the diagram).

c) Write a scalar expression for the net force on the SYSTEM. (Ex: $F_{net} = F_{ap} - F_f$)

$F_{net,system} = F_x - F_{fA} - F_{fB}$

d) If the applied force is 75 N, determine the acceleration of the system (7.03 m/s²)

$F_{net.Ay} = F\sin 25° + F_N - F_g = 0$

$F_N = F_g - F\sin 25°$

$F_N = m_A g - F\sin 25°$

$F_{net.By} = F_N - F_g = 0$

$F_N = F_g \quad F_N = m_B g$

$\vec{F}_{netx} = ma$

$F_{netx} = F_x - F_{fA} - F_{fB}$

$F_{netx} = F\cos 25° - \mu(m_A g - F\sin 25°) - \mu m_B g$

$F_{netx} = 67.9731 - 4.3384 - 7.3575$

$F_{netx} = 56.2772$ N

$a = 56.2772/8 = 7.03465$ m/s² ≈ **7.0 m/s²**

e) Determine the magnitude of the tension in the string between the boxes. (28.5 N)

$F_{net.Bx} = F_T - F_{fB}$

$F_{net.Bx} = m_B \times a$

$m_B \times a = F_T - F_{fB}$

$m_B \times a = F_T - (\mu)(m_B)(g)$

$F_T = (m_B \times a) + (\mu)(m_B)(g)$

$F_T = 28$ N

f) How much can the applied force increase before block A will start lifting off the ground?

$F_{net,Ay} = F\sin 25° + F_N - m_A g = 0$

$F_N = 0$

$F\sin 25° = m_A g$

$F = \frac{m_A g}{\sin 25°}$

$F = 116$ N

$\Delta F = F_{new} - F_{old}$

$\Delta F = 116 - 75$

$\Delta F = 41$ N

Homework

Txt pg. 69 #9 pg 68 #3, pg. 76 #5, pg. 81 #1, pg. 82 #4, 5, pg. 83 #5 pg. 89 #3, 5, 6, 7

Additional Problems

1. A child pulls a 7.0 kg wagon along a flat sidewalk with 55 N of force at an angle of 20.0° to the horizontal. If the co-efficient of friction is 0.35, determine the acceleration of the wagon. [4.9 m/s^2]

2. A shopping cart is being pushed along the aisle of a grocery store with an acceleration of 0.12 m/s^2. The force is being applied along the handle of the cart at an angle of 30° below the horizontal. If the cart and contents have a mass of 12 kg and the co-efficient of friction is 0.35, determine the magnitude of the applied force. [62 N]

3. A train consisting of two cars pulled by a locomotive experiences an acceleration of 0.33 m/s^2 [fwd]. Friction is negligible. Each car has a mass of 3.1×10^4 kg.
    a. Determine the force exerted by the first car on the second car. (1.0×10^4 N)
    b. Determine the force exerted by the locomotive on the first car. (2.0×10^4 N)

4. Two books are resting side by side, in contact, on a desk. An applied horizontal force of 0.58 N causes the books to move together with an acceleration of 0.21 m/s^2 horizontally. The mass of the book to which the force is applied directly is 1.0 kg. Neglecting friction, determine,
    a. The mass of the other book. (1.8 kg)
    b. The magnitude of the force exerted by one book on the other. (0.37 N)

5. A figure skater of mass m = 56 kg pushes horizontally for 0.75 s with a constant force against the boards at the side of a skating rink. Having started from rest, the skater reaches a maximum speed of 75 cm/s. Neglecting friction, determine the magnitude of,
    a. The (constant) acceleration (1.0 m/s^2)
    b. The force exerted by the skater on the boards (56 N)
    c. The force exerted by the boards on the skater (56 N)
    d. The displacement of the skater from the boards after 1.50 s (0.84 m)

6. Two boxes, one with a mass of 60 kg and the other with a mass of 90 kg, are in contact and at rest on a smooth surface. An 800 N force is exerted on the 60 kg box toward the 90 kg box. Calculate:
    a. the acceleration of the boxes. (5 m/s^2)
    b. the magnitude of the action and reaction forces between the boxes. (500 N)

7. A 55 kg girl, Bonnie, facing a 70 kg boy, Gerald, on a frictionless surface pushes Gerald with a force of 250 N for 0. 10 s. Calculate
    a. Bonnie and Gerald's acceleration. (4.5 m/s^2 & -3.6 m/s^2)
    b. Bonnie and Gerald's speed at the end of 0.50 s. (2.3 m/s & -1.8 m/s)